



Semester 1 Examinations 2019 / 2020

Exam Code(s) 1MF1, 1SD1, 1SD3
Exam(s) Master's in Software Design & Development
Higher Diploma in Applied Science (Software Design & Development)

Module Code(s) CT870
Module(s) Internet Programming

Paper No. 1

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Instructions: Answer any 3 questions. All questions carry equal marks.

Duration 2hrs
No. of Pages 3 (Including cover page)
Discipline Computer Science

Requirements None

Question 1:

- a) Explain the steps involved in a browser requesting and displaying a simple HTML webpage, located on a remote server.
(4 marks)
- b) Write HTML that creates an ordered list of 12 elements representing the 12 months of the year (January, February, March...)
(4 marks)
- c) Write HTML that uses a table to display a single special offer product for a fictional e-commerce site. You may choose any product you like. A header cell should be used to contain the title, description, price, and image (the src attribute can contain a mocked link) of the product. Standard cells should be used to display the data about the single item instance. Using the colspan attribute ensure that the description spans a width of two columns.
(6 marks)
- d) Do all HTML tags have a matching closing tag? Give a reason for your answer.
(2 marks)
- e) Create a simple menu using HTML consisting of an unordered list with 3 hyperlinks. The hyperlinks can link to any resource of your choosing.
(4 marks)

20 Marks

Question 2:

- a) Explain how the grid system in Bootstrap works. In your answer discuss the purpose of rows and columns and describe how the grid facilitates responsive design.
(5 marks)
- b) Provide the HTML and corresponding Bootstrap classes to split a row into two separate sections, each sized 6 columns wide on a large device and medium device. In the case of smaller devices, the columns should stack on top of each other.
(5 marks)
- c) In the context of style classes, explain what class and id-based selectors are. Give an example of each.
(5 marks)
- d) In CSS describe using an example how a single style can be applied to all tags of a particular type on the web page.
(5 marks)

20 Marks

Question 3:

- a) Using JavaScript, write code to iterate through an array of objects (called myArray), removing objects placed at **odd** index positions in the array. Assume the array contains 8 vehicle objects which take the following form ({**make**: "Ford", **model**: "Focus", **age**:2}, ... , {**make**: "Toyota", **model**: "Yaris", **age**:5}). You may use the splice method to remove the objects from the array, however each time splice is called it will re-index the array. To counter this you should iterate through the array in reverse. (9 marks)
- b) The splice method can be used to remove items from an array. Considering the above array which contains 8 vehicle objects, write code to show how the splice method can be used to delete the last three objects in the array. (3 marks)
- c) Write JavaScript code that upon button click reads 3 numbers from 3 text boxes and doubles each number before saving it in the text box it came from. (8 marks)

20 Marks

Question 4:

You have been asked to build a NodeJS (Express) backend, which comprises a collection of RESTful APIs for an Instagram style application. Each image shared by the users of the application should contain a description of the image, the upload time and date, the number of likes and the number of re-shares.

- a) Provide a sample JSON document which contains data for a single image shared on the platform. (4 marks)
- b) Write either a MongoDB or Mongoose query (whichever you prefer) which returns all images posted, sorted in **ascending** order by upload time. (5 marks)
- c) Define a server-side API route accessible at the path '/uploadImages' which accepts POST requests. Assume that request body contains JSON data and the payload is processed by the Express middleware and made available as an object in the request body. Either using the object modelling tool Mongoose or a traditional MongoDB query, provide code to save the data in the request body to the MongoDB database. If using Mongoose, you do not need to specify the schema, you may assume a schema named Image is already defined. (7 marks)
- d) Define a server-side API to delete documents from the collection by parsing the request parameters, extracting the id and removing the document found. (4 marks)

20 Marks